

Vandit U Shah

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Summary

AI/ML Engineer with 3+ years of experience designing, developing, and deploying scalable machine learning and data solutions. Skilled in NLP, generative AI, model optimization, and real-time data pipelines. Experienced in cloud platforms, MLOps, and deploying high-availability AI services with Python, SQL, Spark, Kubernetes, and AWS/Azure/GCP. Adept at delivering actionable insights, optimizing models for performance and explainability, and collaborating cross-functionally to drive data-driven decision-making.

Technical Skills

- Programming, Data & Distributed Processing:** Python (PyMuPDF, Pydantic, pandas, spaCy, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, LSTM, ARIMA), SQL, SQLAlchemy, PostgreSQL, MySQL, Hive, PySpark, Spark Structured Streaming, Hadoop
- Machine Learning, NLP & Generative AI:** Hugging Face Transformers (DeBERTa-V3, Longformer, FinBERT, RoBERTa), Mistral, LoRA, QLoRA, PEFT, RAG, LangChain, FAISS, NLU, multi-task classification, anomaly detection, Autoencoders, reinforcement learning, transfer learning, ensemble learning, self-supervised learning, few-shot/zero-shot learning, synthetic data generation, adversarial training, continual learning, NAS, AutoML
- Model Optimization, Tuning & Explainability:** Optuna, Bayesian optimization, k-fold cross-validation, hyperparameter tuning, SHAP, LIME, Explainable AI (XAI), model drift detection, Evidently AI
- Data Engineering & Streaming Architectures:** ETL pipelines, data ingestion workflows, schema design, feature engineering, real-time data processing, Snowflake, Delta Lake, AWS Kinesis, AWS Glue, Azure Databricks, Airflow, vector embeddings, feature stores
- MLOps, Deployment & APIs:** Docker, Kubernetes, AWS EKS, Azure Kubernetes Service (AKS), KServe, Kubeflow, MLflow, CI/CD pipelines, REST APIs, gRPC, Kafka, AWS Lambda, Vertex AI, AWS Bedrock, Azure ML, model monitoring (Prometheus, Grafana), Streamlit, Plotly
- Cloud, Security & Collaboration:** AWS, GCP, Microsoft Azure, AWS Bedrock, Vertex AI, Azure ML, Git, GitHub, GitLab, internal Git repositories, SOC2, PCI-DSS, HIPAA, IT risk controls, cross-functional collaboration, stakeholder communication, documentation, ethical AI practices

Professional Experience

AI/ML Engineer, Plaid Inc.

06/2025 – Present | Remote, USA

- Engineered the SmartTransaction AI platform, an end-to-end transaction intelligence and fraud detection system processing millions of daily banking transactions. Reduced fraud by 32% and improved real-time risk decision latency by 45% across payments and identity verification workflows.
- Constructed scalable end-to-end data pipelines with Python, SQLAlchemy, and Pydantic, normalizing multi-institution financial data into PostgreSQL and feature stores with 99.7% schema validation accuracy, strengthening reliability for internal and developer-facing APIs.
- Designed advanced transaction enrichment and categorization models using NLP (spaCy, FinBERT) to extract merchant descriptors, recurring patterns, and anomalies, increasing transaction classification precision by 20% and enhancing downstream underwriting effectiveness.
- Optimized transformer models (Longformer, FinBERT) on 250K+ labeled financial transactions, boosting fraud detection F1-score from 74% to 89%, while reducing inference costs by 38% with LoRA and PEFT techniques for production scalability.
- Implemented real-time Retrieval-Augmented Generation (RAG) systems with LangChain, FAISS, and AWS Lambda to deliver intelligent financial insights, increasing investigation accuracy by 28% and decreasing manual review time by 35%.
- Integrated agentic AI workflows via AWS Bedrock for anomaly investigations, identity risk scoring, and adaptive fraud mitigation, collaborating cross-functionally to ensure SOC2, PCI-DSS compliance, CI/CD deployment, monitoring, explainability, and high-availability ML services.

AI/ML Engineer, S&P Global

01/2021 – 07/2023 | Ahmedabad, India

- Developed an Intelligent Financial Analytics Platform to process and analyze large-scale market and transaction datasets, improving operational efficiency by 22% and enabling faster, data-driven business decisions.
- Built scalable ETL pipelines using Azure Databricks, Delta Lake, and Spark Structured Streaming to integrate structured financial data with unstructured market insights, accelerating analytics and predictive modeling for investment and risk teams.
- Designed and deployed ML models using LightGBM, Transformer-based Autoencoders, and scikit-learn for market trend forecasting, anomaly detection, and classification of financial instruments, achieving 89% F1-score and 94% recall.
- Applied RoBERTa and Sentence-BERT embeddings with logistic regression to extract actionable insights from financial reports, analyst notes, and transaction logs, reducing manual review workload by 21% and enhancing decision support for research teams.
- Optimized model performance using Optuna for Bayesian hyperparameter tuning and applied SHAP and LIME explainability, ensuring transparency and trust in predictive insights delivered to stakeholders.
- Containerized models with Docker and deployed on Azure Kubernetes Service (AKS) with KServe, enabling real-time analytics and integration into internal dashboards for monitoring and strategic decision-making.
- Implemented continuous training and monitoring pipelines using MLflow, Kubeflow, and Evidently AI, maintaining model accuracy above 91% and supporting reliable, scalable, and automated market analytics operations.

Education

Loyola Marymount University – Los Angeles, CA, USA

Master of Science in Computer Engineering

08/2023 – 05/2026

Indus University – Ahmedabad, India

Bachelor of Technology in Computer Science

08/2018 – 05/2022

Projects

Virtual Event Management System

- Developed a React.js and Node.js platform integrating Stripe payments and Zoom conferencing, enabling seamless registration, attendance tracking, real-time communication, and interactive features, boosting attendee engagement by 50% and supporting scalable large-scale virtual events.

Smart Scheduling & Booking System

- Built an AI-driven calendar booking system using Node.js, React, and PostgreSQL, with multi-vendor support, Stripe payment integration, and Twilio notifications, providing smart time slot suggestions and seamless customer interaction for efficient scheduling and bookings.